

Matthew De Jesus

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EDUCATION

Massachusetts Institute of Technology (MIT)

Cambridge, MA

Master of Engineering (MEng, 1-Year Accelerated Program)

Expected May 2027

- Completed B.S. in Electrical Engineering and Computer Science (GPA: 4.0) — May 2026
- **Relevant Coursework:** Digital Systems, Electrical Circuits, Power Electronics, Computation Structures, Microcomputers, Semiconductors, Nanoelectronics, Computer Systems Engineering, Fundamentals of Programming, Algorithms and Data Structures, Low-level Programming in C and Assembly

EXPERIENCE

Power Integrations

Jun 2026 – Present

GaN Characterization Intern

San Jose, CA

- Assessed reliability of next-generation GaN power devices through Time-Dependent Dielectric Breakdown (TDDB) testing.
- Compared converter operation in discontinuous- versus continuous-conduction modes (DCM vs. CCM).
- Characterized dynamic on-resistance and output capacitance (C_{oss}) losses at high switching frequency.

MIT Research Laboratory of Electronics

Jun 2025 – December 2025

Undergraduate Researcher

Cambridge, MA

- Designed and iterated multiple high-frequency PCB test platforms to characterize switching loss, dynamic on-resistance, and output capacitance losses in 5th generation GaN FETs.
- Automated laboratory instrumentation using SCPI (power supplies, oscilloscopes, thermal control) to execute voltage, frequency, and temperature sweeps for reproducible device characterization.
- Co-authored paper accepted to the 2026 IEEE Applied Power Electronics Conference; to appear in IEEE Xplore.

MIT Energy Initiative

Jun 2024 – Aug 2024

Undergraduate Researcher

Cambridge, MA

- Built a simulation to model energy arbitrage in the Texas power market, balancing speed and estimation accuracy.
- Used a pruning approach to keep only the top N profitable paths each hour, reducing computation while focusing on the most profitable outcomes; identified the highest-income scenario and generated clear revenue and ROI summaries.
- Added robust input checks and error handling for user-selected parameters (load zones, file types, etc.).

Bloomberg Summer Internship

Jun 2023 – Aug 2023

Intern, Employee Operations

New York, NY

- Managed computers on the Bloomberg network using Active Directory, including remote installation of programs.
- Enrolled laptops into Microsoft Azure, ensuring seamless integration and enhanced functionality.
- Led Scrum projects using the Agile framework, driving team collaboration and efficient project completion.

PROJECTS

Differential Amplifier IC (Fabricated by Intel)

Feb 2025 – May 2025

- Designed a CMOS differential amplifier in Intel 22nm PDK achieving 80 dB gain, 1 mW power, 200 mV swing.
- Implemented inverter, passgate, and digital code checker for integration into shared bus IC architecture.
- Verified performance in Cadence (gain, swing, power, node voltages) and delivered DRC/LVS-clean submission.
- Tuned transistor sizing and biasing to maximize figure of merit, balancing gain, swing, and power.

Go-Cart Motor Drive

Aug 2024 – Sept 2024

- Designed a PWM signal with adjustable duty cycle and frequency using a 555 timer and a 311 comparator.
- Integrated delay circuitry to control two N-channel MOSFETs in a totem-pole configuration for efficient switching.
- Achieved high-power handling capability with 36V and 30A, successfully driving a 1.5HP motor for a go-kart.
- Demonstrated expertise in PWM motor control and power electronics, managing component selection and cost.

RISC-V Processor

Aug 2023 – Dec 2023

- Designed and implemented a 32-bit RISC-V 5-stage pipelined processor with a 600 ns critical path.
- Integrated a 32-bit direct-mapped cache and a 2-way, 32-bit associative cache to enhance processor efficiency.
- Optimized the processor by implementing branch prediction and full bypassing from writeback and execute.

SKILLS

- **Programming Languages:** Verilog, Python, C, Assembly, JavaScript, Bluespec, LISP (Scheme), PSOC, FPGA
- **Electrical:** Networking, Cable Routing, Soldering, Electronics Repair, Computer Building, Power Electronics